



# PaN-Finder Deliverable 7.b:

## Users Interviews

### Table of Contents

|                                            |    |
|--------------------------------------------|----|
| 1 Introduction.....                        | 2  |
| 2 Interview Structure.....                 | 2  |
| 2.1 Introduction.....                      | 2  |
| 2.2 Feedback collection.....               | 2  |
| 2.3 Closing remarks.....                   | 2  |
| 3 User 1.....                              | 3  |
| 3.1 Example Searches.....                  | 4  |
| 3.1.1 First Search.....                    | 4  |
| 3.1.2 Second search.....                   | 4  |
| 3.1.3 Third search.....                    | 5  |
| 3.1.4 Fourth search.....                   | 6  |
| 3.1.5 Fifth search.....                    | 7  |
| 3.1.6 Sixth search.....                    | 8  |
| 3.1.7 Seventh search.....                  | 9  |
| 4 User 2.....                              | 10 |
| 5 Future Improvements 1.....               | 10 |
| 6 Future improvements 2.....               | 12 |
| 7 User 3.....                              | 12 |
| 8 Future Improvements 3.....               | 12 |
| 9 User 4.....                              | 13 |
| 9.1 Example Search and Results Review..... | 14 |
| 10 Future Improvements 4.....              | 15 |
| 11 User 5.....                             | 16 |
| 12 User 6.....                             | 17 |
| 13 User 7.....                             | 18 |
| 14 User 8.....                             | 19 |
| 15 Funding.....                            | 20 |
| 16 Licensing.....                          | 20 |



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# 1 Introduction

This deliverables contains the feedback received from users that kindly gave their availability to be interviewed. It also contains the action items that have been decided by the project team following the user interview or groups of them.

At the moment of this writing, the project team has interviewed 8 users which provided a wide range of feedback. For each users, we have reported the type of user and feedback as a list of unordered comments or thoughts that the user has shared with us and are relevant in the context of the project. During the interview, the interviewer writes down the user feedback and thoughts. Post interview the notes are transcribed and summarized to be added here and also to guide future development of the project.

# 2 Interview Structure

Each interview is structured in three main sections:

## 2.1 Introduction

The participant is informed about the purpose of the project and what is the feedback is sought. If the participant is not familiar with the PaN-Finder, few examples are provided.

The participant is directed to focus on three main areas: scientific relevancy of the results returned by the PaN-Finder, UI/UX (user interface and user experience), natural language interaction.

## 2.2 Feedback collection

This section of the interview is lead as an unstructured conversation with the focus to capture all the information that the participant is willing to share. The interaction is kept informal and open in order to let the participant be at ease to provide feedback free of biases due to perceived judgments from the interviewer. When possible the interviewee led the discourse with minimal intervention from the interviewers. In some interviews, the interviewers had to stimulate and/or focus the conversation.

## 2.3 Closing remarks

A short time where the interviewers summarized the main points of the feedback in order to confirm their understanding. The interviewee is also asked to share their final remarks on the tool, the project, potential and innovative way of use it, improvements and last thoughts.



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### 3 User 1

*Date:* 2025/09/23

*User type:* domain expert, power user

*Location:* in person

*Collected feedback:*

- Remove the LLM prompt above the table. Not informative.
- Provide a relevance summary (why the item is important in the context of user query) on demand for each item.
- For each row of the table, add a "summary" button which loads and show the relevance summary for the selected item.
- Automatically show the relevance summary for the first item, which is the most relevant
- For each item provide more information regarding the relevance.

Suggested solution: show the item relevance score on the item row all the way to the right.

Color the background of the html element containing the score with a continuous color scale going from green for relevancy 1 to red for relevancy 0

- Do not show item raw data by default. Only if the user expand the element, show the raw data.
- Add explanation to thumb up/down for feedback.  
It should read something like: Is this result relevant to your query?
- Add a banner at the bottom of the page to encourage the user to rate the application as a whole.
- It might be useful to implement a ChatGPT experience, but we are not sure about how much added value it would bring. First we need to improve the over all accuracy of the app.
- An easier improvement could be a item follow-up: allow the user to ask follow-up questions on a specific item.
- User checked at most 5 results when providing feedback on relevancy
- Suggestion about the output:

- indicate year of data collection and instrument(s). See last comment.
- explicitly explain how the results are ordered, which is by relevance

- Add a feedback form. Scout the web for examples of such forms. One example is shown in figure 1.
- The comments at the bottom should be displayed in a different way. It is a bit difficult to see that it is not a description of one of the results. See screenshot in figure 2.



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An OSCARS Project, Funded by the European Union

✕

If a colleague asked about your experience using this software, how would you rate it on a scale of 0 to 10?

Poor

0

1

2

3

4

5

6

7

8

9

10

Excellent

Figure 1: Example of feedback form

- For the instruments, a simple list should be enough. Example:
  - (ILL) D4, IN5
  - (ESS) DREAM, LOKI
  - (MaxIV) DANMAX
- The same should be done for the years with the option of indicating the range for consecutive years (2015-2017).
- I find that giving only the facility is not enough since it does not give any hints on the scattering techniques: imaging, spectroscopy, sans, saxs, reflectometry, diffraction, spin-echo,...
- Having the instruments listed narrows the range of options
- I would provide this info even if the data is under embargo.

## 3.1 Example Searches

### 3.1.1 First Search

**Query:**

Find datasets collected at a neutron facility where the sample is chalcogenide

**Results:**

Most Directly Related Result is the following:

The dataset titled Test samples **fully aligns with** your request, as it was collected at a neutron facility and directly involves chalcogenide samples, making it especially pertinent to your search. (DOI: 10.5291/ILL-DATA.TEST-2357)

### 3.1.2 Second search

**Query:**

Find datasets collected at D4 at ILL where the sample is chalcogenide

**Results:**

All the results listing instruments other than D4 are not relevant



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### 3.1.3 Third search

#### Query:

Find datasets collected at D4 at ILL in 2015 where the sample is chalcogenide

#### Results:

No specific comments are provided.

Screenshot of part of the results

Relevance Explanation

**Relevant Results**

- The study *Unraveling the structural ordering of water in amorphous calcium carbonate using neutron pdf-analysis* **fully aligns with** your request by presenting data collected at the D4 instrument at ILL in 2015, though it focuses on calcium carbonate rather than chalcogenide samples. (DOI: [10.5291/ILL-DATA-6-06-464](https://doi.org/10.5291/ILL-DATA-6-06-464))
- The document *Investigating the temperature evolution of EuTiO3 local structure by means of Pair Distribution Function Analysis* **directly satisfies all aspects of** your search criteria regarding instrument, facility, and year, but the sample studied is EuTiO3, not a chalcogenide. (DOI: [10.5291/ILL-DATA-5-23-669](https://doi.org/10.5291/ILL-DATA-5-23-669))

**Additional Background & Context**

- The entry *Test samples* provides minimal information and does not specify sample composition, but may offer general context about experiments conducted at ILL. (DOI: [10.5291/ILL-DATA-TEST-2357](https://doi.org/10.5291/ILL-DATA-TEST-2357))
- The article *Structural transformation in water at high pressure* presents neutron diffraction studies on water under varying conditions at ILL, which may be tangentially relevant for understanding experimental techniques but does not address chalcogenide samples. (DOI: [10.5291/ILL-DATA-6-02-547](https://doi.org/10.5291/ILL-DATA-6-02-547))

These results indicate that while several datasets were collected at D4 at ILL in 2015, none specifically involve chalcogenide samples as requested. The most relevant documents meet your criteria for instrument, facility, and year, but focus on other materials, offering insight into the types of studies conducted and the experimental approaches used. The additional documents provide broader context about research activities at ILL, though they do not directly address your query.

Figure 2: User 1 search 2 results



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### 3.1.4 Fourth search

#### Query:

find datasets collected with polarised neutrons using Helium 3 cells

#### Results:

The results show in the top part of figure 3 are fine.

#### Relevant Results

- The proposal *Investigation of high pressure phase in multiferroic perovskite DyMnO<sub>3</sub> by using CryoPAD and Hybrid-Anvil-Cell* describes experiments involving spherical neutron polarimetry under high pressure, which **addresses several key aspects** of your query by utilizing polarized neutrons and specialized cells, though it does not specify the use of Helium 3 cells. (DOI: [10.5291/ILL-DATA.5-54-293](https://doi.org/10.5291/ILL-DATA.5-54-293))
- The study *Polarized Neutron Scattering Study on Q1D superconductor K<sub>2</sub>Cr<sub>3</sub>As<sub>3</sub>* reports on half-polarized neutron diffraction measurements in superconductors, **covers part of what you're looking for** by focusing on polarized neutron techniques, but does not mention Helium 3 cells. (DOI: [10.5291/ILL-DATA.5-51-517](https://doi.org/10.5291/ILL-DATA.5-51-517))
- *Polarized neutron study on the multipolar component in a quasi-doublet quantum magnet* explores magnetic excitations using polarized neutron beams, **addresses several key aspects** relevant to your interest in polarized neutron experiments, though it does not specify the use of Helium 3 cells. (DOI: [10.5291/ILL-DATA.4-05-755](https://doi.org/10.5291/ILL-DATA.4-05-755))

#### Additional Background & Context

- The paper *Spin density distribution on Ru in Ca<sub>2</sub>RuO<sub>4</sub>: A polarised neutron diffraction study* investigates spin density distributions using polarized neutron diffraction, offering background on the application of polarized neutrons in material studies, though it does not reference Helium 3 cells. (DOI: [10.5291/ILL-DATA.5-51-496](https://doi.org/10.5291/ILL-DATA.5-51-496))
- *Structural characteristics of bioinspired amorphous calcium carbonate: Total scattering using polarized neutrons* provides insight into the use of polarized neutron scattering for studying structural water in biominerals, contributing contextual information about polarized neutron techniques without mention of Helium 3 cells. (DOI: [10.5291/ILL-DATA.6-05-980](https://doi.org/10.5291/ILL-DATA.6-05-980))

These results collectively highlight datasets and studies that involve polarized neutron techniques and specialized cells, which are closely related to your query. However, none of the documents explicitly mention datasets collected with polarized neutrons using Helium 3 cells, though several address related experimental approaches and may offer useful context or partial relevance.

Figure 3: User 1 Search 4 results



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Report on current system architecture

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### 3.1.5 Fifth search

#### Query:

datasets of magnetostrictive galfenol thin films using x-rays or neutrons.

#### Results:

The results show in the top part of figure 4 are fine.

#### Relevance Explanation

##### Relevant Results

- The paper *Magnetic ordering at the Verwey transition in bulk like Fe<sub>3</sub>O<sub>4</sub> thin films* explores magnetic properties and thin film growth techniques using advanced characterization, which **addresses several key aspects** of your interest in magnetostrictive thin films studied with neutron or x-ray methods, though it does not focus on galfenol specifically. (DOI: [10.5291/ILL-DATA.5-54-203](#))
- The study *Neutron reflectometry of exotic epsilon-Fe<sub>2</sub>O<sub>3</sub> thin films* investigates magnetic and electric properties of iron oxide thin films using neutron reflectometry, **covers part of what you're looking for** by demonstrating relevant experimental techniques, but does not involve galfenol or magnetostriction directly. (DOI: [10.5291/ILL-DATA.5-54-226](#))

##### Additional Background & Context

- The article *Magnetic structure determination in o-TbMnO<sub>3</sub> and o-LuMnO<sub>3</sub> thin films by combining both neutron diffraction and spherical neutron polarimetry* discusses the use of neutron-based techniques to study magnetic structures in thin films, offering methodological insights that may be useful for your research on magnetostrictive materials. (DOI: [10.5291/ILL-DATA.5-54-153](#))
- The study *Magnetic structure of room temperature multiferroic thin film BiFeO<sub>3</sub>* examines the magnetic properties of multiferroic thin films at room temperature, providing general background on thin film magnetism and characterization approaches relevant to your broader topic. (DOI: [10.5291/ILL-DATA.5-41-695](#))

These results do not directly provide datasets on magnetostrictive galfenol thin films studied with x-rays or neutrons, but they offer related studies on thin film magnetism and experimental techniques that may inform your search or methodology.

Figure 4: User 1 Search 5 results





## PaN-Finder Deliverable 7.b

Report on current system architecture

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### 3.1.6 Sixth search

**Query:**

datasets of pdf time-of-flight neutron experiments

**Results:**

I don't think that my query was understood. I did not get any results related to PDF experiments

#### Relevance Explanation

##### Relevant Results

- The study *New Neutron Electric Dipole Moment Search using a Beam* describes experimental work with pulsed neutron beams, which is pertinent to your interest in time-of-flight neutron experiments and may offer useful insights into experimental setups and data collection. (DOI: [10.5291/ILL-DATA.3-07-374](https://doi.org/10.5291/ILL-DATA.3-07-374))
- *Use of TOF analysis of pulsed, monochromatic neutrons to suppress incoherent scattering from 1H containing materials* presents the development and validation of a powder diffractometer using time-of-flight analysis, **addresses several key aspects** of your query by discussing experimental approaches and data relevant to time-of-flight neutron techniques. (DOI: [10.5291/ILL-DATA.1-10-24](https://doi.org/10.5291/ILL-DATA.1-10-24))
- The paper *Probing excess oxygen impacts on lattice dynamics of Nd<sub>2</sub>NiO<sub>4</sub>+d using neutron time-of-flight spectroscopy* investigates material properties using neutron time-of-flight spectroscopy, providing experimental data and analysis methods that are closely related to your search for datasets from such experiments. (DOI: [10.5291/ILL-DATA.7-01-373](https://doi.org/10.5291/ILL-DATA.7-01-373))

##### Additional Background & Context

- *Transport of Water Across Liposome Membranes Studied by Contrast Variation Stopped-Flow Experiments* utilizes neutron scattering in kinetic studies, offering broader context on experimental techniques that may be tangentially relevant to time-of-flight neutron data acquisition. (DOI: [10.5291/ILL-DATA.9-13-591](https://doi.org/10.5291/ILL-DATA.9-13-591))
- The document *Search for new gravity-like interactions at nanometer range using cold neutron beams* explores fundamental physics using cold neutron beams, providing background on experimental neutron studies that, while not directly focused on time-of-flight methods, may inform your understanding of neutron experiment datasets. (DOI: [10.5291/ILL-DATA.3-15-89](https://doi.org/10.5291/ILL-DATA.3-15-89))

Figure 5: User 1 Search 6 results





## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union

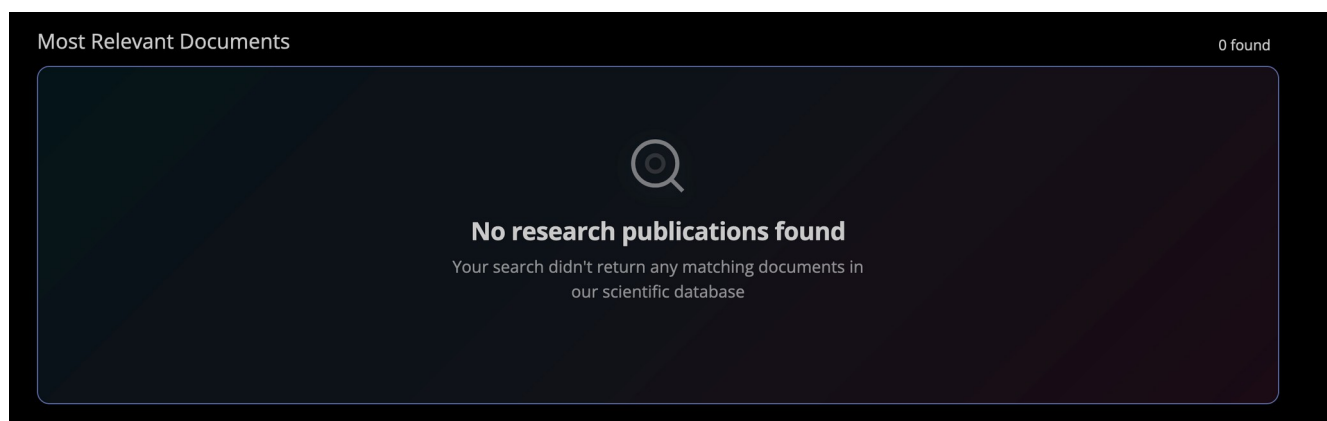
### 3.1.7 Seventh search

**Query:**

find datasets where the same sample was measured at neutron and synchrotron facilities

**Results:**

No results returned



*Figure 6: User 1 Search 7 results*



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

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### 4 User 2

*Date:* 2025/09/26

*User type:* domain expert, power user

*Location:* in person

*Collected Feedback:*

- Queries will concentrated around a specific compound.
- Scientific field: soft matter
- Acronyms are important.  
Maybe the tools should be aware of acronyms in the specific field of science
- Add guidance on how to write queries
- Interactive suggestion while writing user prompt
- The overall relevance summary is valuable
- Leave high / low relevance indication for each item
- Add a checkbox for each item to be selected for the follow-up query
- Individual item raw data is valuable to verify that is actually relevant
- Show raw data in a json expandable widget and not in a scrollable text area
- Remove summary, which is AI generated and we use only for similarity search
- Rename content to Summary/Abstract
- Expose the following information from the raw data:
  - Sample(s)
  - Instrument(s)
  - Chemical formula(s)
  - Technique from PaNET
- Ingest also Lab Report where available

### 5 Future Improvements 1

*Date:* 2025/09/26

*Action items:*

- Remove top LLM summary prompt
- Add LLM relevancy summary to each item
- First item will automatically expanded with relevancy summary
- All other items can be expanded by the user
- The relevancy summary will be generated when item is expanded for the first time,
- Add string "Relevant" right before thumbs up/down to clarify functionality



## PaN-Finder Deliverable 7.b

Report on current system architecture

An OSCARS Project, Funded by the European Union

- Show following info in the same order for each item when expanded:
  - DOI
  - Relevancy summary
  - Abstract/Summary
  - Raw Data
- Raw Data should be collapsed and shown with collapsible JSON view
- Update relevancy information on item row with info from "knee point" and "relevancy score".
  - Knee point will be shown as Relevant and Suggested
  - Relevancy score will be shown as percentage
  - Color will be going from green to red according to relevancy score
  - See figure 7 for wireframe
- Add general user feedback at the bottom

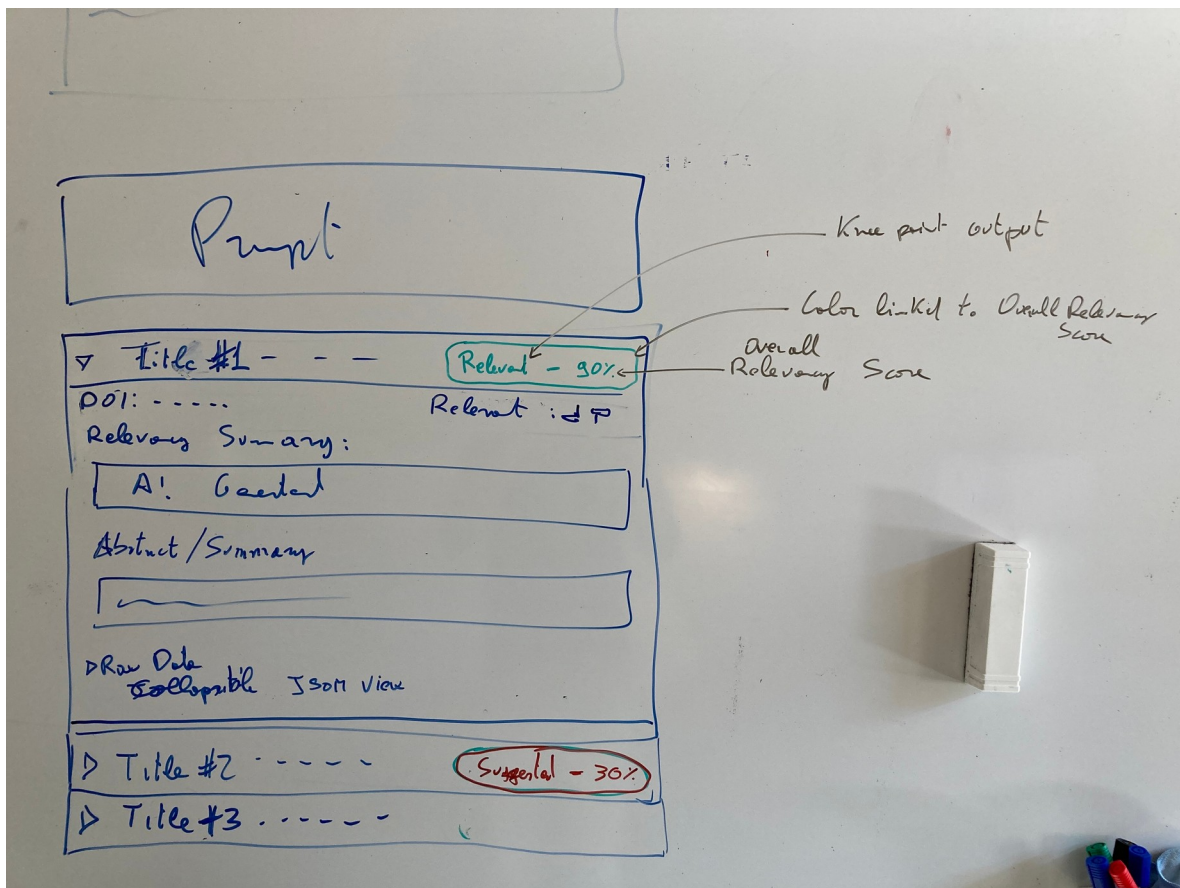


Figure 7: Wireframe design for new UI item element



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union

## 6 Future improvements 2

*Date:* 2025/09/26

*Nice-to-have items:*

- Interactive suggestion while writing the prompt
- Include acronyms in processing
- Relevance/Selection checkbox on each items for follow-up queries

## 7 User 3

*Date:* 2025/10/08

*User type:* domain expert, power user

*Location:* on-line

*Collected Feedback:*

- Field of science: structural biology
- As it is now, PaN-Finder might not be useful because it has no explicit link to Publications or other repositories or dbs, such as PDB
- Example queries:
  - X-ray diffraction about structure of bovine insuline
  - datasets where the publisher is Max-IV and the beamline is BIOMAX
  - Datasets associated with PDB entry 6aR7
  - Datasets associated with PDB entry 4AR4
  - Datasets from ESRF acquired on beamline ID-29
  - Serial synchrotron crystallography
- It is important to consider the synonyms and abbreviations, such as:
  - instrument = beamline
  - Hemoglobin = Haemoglobin = Hb = Hgb
  - Myoglobin = Myb = Mb = MB
- Overall, the frontend is well structured. It provides the correct information while using the available space efficiently

## 8 Future Improvements 3

*Date:* 2025/10/09

*Nice-to-have items:*

- in the field of structural biology, the connection with PDB entry is important.

*Questions*



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

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- should we rely on the curation done by the data provider or should we further curate the data during ingestion?
- If we do additional data curation of the data internally, how should we do it and do it reliably?

## 9 User 4

*Date:* 2025/10/13

*User type:* domain expert

*Location:* email communication

*Collected Feedback:*

- The user provided a written report with comments
- We should take whole text into account in absence of metadata (check for "high energy x-ray" when filter is "energy > 2 keV")
- Prompting XRR excludes NR but shows other x-ray techniques
- Can limit to one facility/publisher (only ILL)
- Cannot exclude one publisher (Not ILL)
- PaN-finder enables more flexible search
- LLM gives "smart" responses why results fit prompt
- Will admit if results fail specifications
- Result ordering is not well explained
- Great potential for finding FAIR data
- Will benefit from other projects related to extracting metadata
- Make PaN-finder easier to find
- Ideas for result quality:
  - More weight to specified technique
  - Ensure top result meet search specifications
  - Show data availability in answer, examples:
    - Embargoed until <date>
    - Data available
  - Automatically search for publications connected to proposal ID
- Acronyms used:
  - **XRR** = x-ray reflectometry
  - **NR** = Neutron reflectometry
  - **GIXD** = grazing-incidence x-ray diffraction
  - **GIWAXS** = grazing-incidence wide-angle x-ray scattering (can be often used interchangeably with GIXD)



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- The user provided a document with the results from query "X-ray reflectometry with photon energy > 2 keV" properly labeled with his comments. See dedicated section 9.1
- User suggested to add button to download results in a csv format

## 9.1 Example Search and Results Review

User number 4 provided a detailed review of the results that the system returned when he submitted the following query:

*X-ray reflectometry with photon energy > 2 keV*

The results and the user review is reported in the following table.

In their review, the user also included the technique used to acquire the data. The technique expressed as structured property is missing for almost the entirety of the body knowledge. It is a future goal of the PaNOSC community to properly tag every public data available with the correct technique and make sure that such technique is present in the PaNET ontology.

| User feedback |             | DOI<br>Title                                                                                                                                                                             | Facility | Relevancy<br>Score (%) |
|---------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------|
| Relevance     | Technique   |                                                                                                                                                                                          |          |                        |
| Low           | SAXS like   | 10.16907/c4fdaf60-562b-4501-a314-a7c153d676ee<br>X-ray scattering tensor tomography data for a validation sample                                                                         | PSI      | 53                     |
| None          | Tomography  | 10.16907/b0e8ef2b-836a-4e29-90bb-38bfebe26685<br>Synchrotron radiation X-ray tomographic microscopy datasets for Early Cretaceous Hedyosmum-like fossils from Portugal and North America | PSI      | 43                     |
| None          | Diffraction | 10.48391/cff474da-2b3f-4a3a-9226-9c4f1d62470<br>[SE] FragMAX screening of SARS CoV-2 nsp10-nsp14ExoN complex                                                                             | MaxIV    | 43                     |
| None          | Tomography  | 10.16907/4d7143fa-21c4-4f3c-8e62-0ffa1720b2ae<br>Synchrotron radiation X-ray tomographic microscopy datasets for Early Cretaceous mosses from the Catefica mesofossil flora, Portugal    | PSI      | 43                     |
| None          | Diffraction | 10.16907/808de0df-a9d3-4698-8e9f-d6e091516650<br>JUNGFRAU detector for brighter X-ray sources - solutions for IT and data science challenges in macromolecular crystallography           | PSI      | 43                     |
| None          | Diffraction | 10.16907/d699e1f7-e822-4396-8c64-34ed405f07b7<br>Synchrotron Imaging of Complex Vascular Lesions in Human Pulmonary Hypertension: Pathology Distribution in 3D Space                     | PSI      | 43                     |
| None          | Tomography  | 10.16907/05a50450-767f-421d-9832-342b57c201af<br>Selection of representative datasets for data compression investigations                                                                | PSI      | 43                     |





## PaN-Finder Deliverable 7.b

Report on current system architecture

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|      |                       |                                                                                                                                                                                                          |       |    |
|------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|
| None | Diffraction           | 10.48391/91a3a886-df6f-434a-9a1d-7304a0061e00<br>Remote experiments at BioMAX                                                                                                                            | MaxIV | 43 |
| None | Diffraction           | 10.48391/b0c36bb8-a00c-4519-8dcc-08d5ca60a313<br>Reference Lysozyme data from the first kilohertz serial crystallography experiments with the JUNGFRU 4M detector at a 4th generation synchrotron source | MaxIV | 43 |
| None | Tomography            | 10.16907/7eb141d3-11f1-47a6-9d0e-76f8832ed1b2<br>Micrometer-resolution X-ray tomographic imaging of a complete intact post mortem juvenile rat lung                                                      | PSI   | 43 |
| None | Tomography            | 10.16907/ef0cfe43-0525-4e80-b77e-d6c979b6acca<br>Airspace diameter map - a quantitative measurement of all pulmonary airspaces to characterize structural lung diseases                                  | PSI   | 43 |
| High | X-ray Reflectometry   | 10.15151/ESRF-ES-1824853946<br>Toward X-ray reflectivity measurements of $\mu\text{m}$ -sized samples                                                                                                    | ESRF  | 43 |
| High | X-ray Reflectometry   | 10.15151/ESRF-ES-874830408<br>Probing interfaces with X-ray reflectivity                                                                                                                                 | ESRF  | 38 |
| Low  | Neutron Reflectometry | 10.5291/ILL-DATA.INTER-463<br>Reflectometry measurements on super-mirrors                                                                                                                                | ILL   | 33 |
| High | X-ray Reflectometry   | 10.15151/ESRF-ES-789629729<br>In Situ GISAXS and XRR Investigation of Microstructural Evolution and Thermal Diffusion in Ion Beam Deposited Fe/Co Nano-Multilayers                                       | ESRF  | 27 |
| High | X-ray Reflectometry   | 10.15151/ESRF-ES-1162136801<br>X-Ray Reflectivity of chips for quantum application                                                                                                                       | ESRF  | 22 |
| Low  | Neutron Reflectometry | 10.5291/ILL-DATA.9-13-488<br>Photoswitchable lipid monolayers                                                                                                                                            | ILL   | 20 |
| None | Tomography            | 10.15151/ESRF-ES-1548496957<br>Imaging of various samples with X-ray nanotomography                                                                                                                      | ESRF  | 18 |
| None | Absorption            | 10.15151/ESRF-ES-496100225<br>X-ray transmission/absorption tests                                                                                                                                        | ESRF  | 18 |
| High | X-ray Reflectometry   | 10.15151/ESRF-ES-2004124768<br>XRR and GID characterization of 2D materials (without reactor)                                                                                                            | ESRF  | 17 |

## 10 Future Improvements 4

Date: 2025/10/17

Action items:

- Labeled results will be used for further training and benchmarking
- We need to review the sub-scores to be able to further train and improve the system
- We need to include the acronyms provided above and a selection of synonyms, like:
  - "energy > 2 KeV" to "high energy x-ray"
- Add csv results download functionality



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- Add if data is accessible or is under embargo
- Use the user feedback to improve training data and benchmarks

## 11 User 5

*Date:* 2025/11/03

*User type:* Domain expert, power user

*Location:* in person

*Collected feedback:*

- Change "raw data" to something like "Entry metadata".  
For scientist "raw data" means sensor data from the file.
- why are we presenting only 20 results.  
Can we provide multiple options to the user and select 20 as default?  
We need to explore the consequences on performance and resources usage.
- At the moment, keywords are mostly single words.  
Can we include the concept of "composite keywords", aka multi words keywords?
- Provide relevance explanation on demand. In most cases, domain expert and expert users do not need the explanation.
- Include a link to the expert mode (editing individual query components) right below the user query.  
It can scroll the page to the bottom to show the current component location or we can transfer the components editing window between query and results adding a expand/collapse capability
- Include examples and how-to for query components
- Add information on Authors/PIs and Instruments and maybe technique to the details of each entry
- Also add sample information if available
- Introduce and expand the synonym concept, where the two words can be synonyms because of meaning, chemical properties or other type of properties  
Example: emin vs C2min vs C2C2im.  
It is author preference which one to use
- Change feedback to collect information:  
which results group does item belong to HR (high relevance), LR (low relevance) or NP (should not be present)



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union

## 12 User 6

*Date:* 2026/01/19

*User type:* Technical expert user

*Location:* on-line

*Collected Feedback:*

- Relevance explanation is useful
- Relevance score is not clear. It might be useful to breakdown the subqueries' scores and explain them
- Likes the query details
- The button to edit the query components is not visible  
It might be better to place it to the right next to the title of the text box
- Location of query components is optimal.
  - It might a better solution to place them between the prompt and the results.
  - Show them collapsed and allow expert users to expand and interact with them
- Change "Raw Data" label to "PaN-Finder Metadata"
  - Scientific users have a different concept for "raw data"
- Relevancy Scores are useful.
  - Maybe make the background color more visible
  - Expand the background color to the whole title cell
- Add explanation of the relevancy score when the item is expanded
- The difference between last result of relevant and first of suggested is minimal.  
That might be confusing
- When searching for entries related to "Munich Crystallography BAG", way to many entries with the same title are shown.
  - We checked with the data source (ESRF) and they are all valid entries
  - How do we differentiate between all of them?
- It is confusing and frustrating when the user clicks on one entry, visit the original page and discover that the data are not accessible because they are still under embargo.
  - Can we provide the embargo information in the entry's details?
- Provide an overall summary covering all the results.
  - Provide such summary upon request by the user
- Track clicks on each title of the entry returned
- User feedback as presented right now (aka on the left of the entry details) is not effective.
- It might be better to explicitly request feedback in 10% of the visits
- Add Release (End of embargo period) and Start Dote in entry details



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union

- Ask explicitly if the entry was use relevant when the user click on DOI and visit the source data page

## 13 User 7

*Date:* 2026/01/30

*User type:* Technical expert user

*Location:* on-line

*Collected Feedback:*

- It would be useful if we can add links to the publications
- It might be useful to develop it as a tool to explore repositories
  - Something like Materials cloud repository
- Information provided for each entry is clear
- Add information about the embargo: when the embargo period ends
- It might be useful to add functionality to order results in different ways:
  - by relevancy (default)
  - by creation date
  - by other columns
- Each group of results (Relevant and suggested) can be reordered individually
- Add publication date both in summary table and entry details
- It might be useful to add file format in ingested data so it can be used by the user in their search prompt.
- Feedback widget is not disturbing the UX
- Regarding the query components:
  - it is clear how to edit the individual components
  - it is not clear how to submit the new search once the user has modify it
  - maybe move search and clear buttons below the text area
  - it is misleading the fact that the original user prompt is not removed when the user is submitting a new query by modifying the query components
- there is no need for query history



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union

### 14 User 8

*Date:* 2026/03/12

*User type:* Domain expert, expert user

*Location:* in person

*Collected Feedback:*

- The user is looking specifically for datasets collected with techniques belonging to the family of GRACING INCIDENCE DIFFRACTION
- This technique has many acronyms and sub techniques
  - GID
  - GIXD
  - GIWAXS Gracing Incidence Wide Angle
  - GIXRD
- First example of relevant user prompt:  
"Gracing Incidence Diffraction on beamline 1010"
- LLM generated relevance explanation should be shorter, concise and high light in bold why it is relevant. It should highlight the words or section why the entry is important.
- The user reported that 15 minutes in person were really useful to better capture the functionalities and learn how to use more effectively the PaN-Finder
- Produce and distribute training materials, such as:
  - Video
  - Web page
  - Few examples with explanation
- It could be useful to offer the user to functionality to change the number of results.
- The number of results requested could also be included and extract from the search prompt
- It is not clear which solution will be more effective:
  - allowing the user to change the number of results with the risk that we return entries with no relevancy at all
  - instructing the user to change or modify their prompt to improve the results set
- Second example of user relevant prompt:  
*I need datasets to train/validate a new model. Can you find at least <n> datasets collected with technique <technique\_name> on instrument <instrument\_name> or an equivalent one?*
- Add feedback on the user prompt or suggestions on how to improve it.
- Add in entry details end of embargo period if information is available.



## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union

### 15 User 9

*Date:* September 26, 2025

*User Type:* Software Expert, Power User

*Location:* By Email

*Collected Feedback:*

- UI Experience:
  - Modern and intuitive design.
  - Clear usability, but the JSON query details may not be immediately understandable for scientific users.
- Query Flexibility:
  - Queries generated from text input feel restrictive.
  - Unclear which data catalogs are being searched (e.g., missing datasets from PSI despite institutes being listed).
  - Need for clarity on whether public/private datasets are included in results.
- Difficulty verifying if all relevant datasets are being returned (e.g., tested searches yielded fewer results than expected).
- Request for better visibility into how data is discovered and filtered.
- JSON query modification feels like a developer feature, not essential for scientific users.
- Purpose of the tool should be to abstract away database query complexities.
- Replace JSON dumps with a more intuitive user-friendly formats.
- It is not clear which databases within the institutes are being queried (e.g., link to specific databases, not just general websites).
- Appreciation for the tool's development and its importance.
- Emphasis on user experience, transparency, and confidence as key factors for regular use.

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## PaN-Finder Deliverable 7.b

*Report on current system architecture*

An OSCARS Project, Funded by the European Union



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